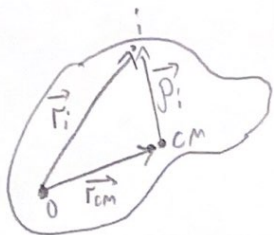


PHYS 302 - HW3

1)



→ We have realized in our earlier studies that total angular momentum of a rigid body w.r.t an inertial frame with it's origin centered at O can be expressed as:

$$\vec{L}_T = M \vec{r}_{cm} \times \vec{V}_{cm} + \sum_{i=1}^N m_i \vec{p}_i \times (\vec{\omega} \times \vec{p}_i)$$

Also $T_{cm}^{of} = \frac{1}{2} \vec{\omega} \cdot \vec{L}_T = \frac{1}{2} \omega_a I_{ab} \omega_b$

Since origin is at CM. Angular momentum "of" the CM is zero since $\vec{r}_{cm} = 0$. Then we're only left with:

$$\vec{L}_T = \sum_{i=1}^N m_i \vec{p}_i \times (\vec{\omega} \times \vec{p}_i) \quad \text{with} \quad \vec{\omega} \times \vec{p}_i = \left(\frac{d\vec{p}_i}{dt} \right)_{inertial}$$

Since our rigid body is free of torque $\Rightarrow \frac{d\vec{L}_T}{dt} = 0$

Since our rigid body is free of force $\Rightarrow \frac{dT_{cm}^{of}}{dt} = 0$

$$(\vec{L}_T)_a = I_{ab} \omega_b \quad (\text{sum over } b)$$

$$\left(\frac{d\vec{L}_T}{dt} \right)_a = 0 = \frac{dI_{ab}}{dt} \omega_b + I_{ab} \frac{d\omega_b}{dt} = 0 \quad \left. \begin{array}{l} \text{Multiply both sides} \\ \text{with } \omega_a \end{array} \right\}$$

$$\omega_a \frac{dI_{ab}}{dt} \omega_b + \frac{d\omega_b}{dt} \underbrace{I_{ab} \omega_a}_{L_b} = \omega_a \frac{dI_{ab}}{dt} \omega_b + \frac{d\omega_b}{dt} \underbrace{I_{ba} \omega_a}_{L_b} \quad (\text{sum over } a)$$

I_{ba} since moment of inertia tensor is symmetric

$$0 = \omega_a \frac{dI_{ab}}{dt} \omega_b + \frac{d\omega_b}{dt} L_b$$

Now since there is no external torque or force acting on the system, rotation kinetic energy $T_{\text{cm}}^{\text{of}}$ will be conserved.

$$\frac{dT_{\text{cm}}^{\text{of}}}{dt} = \frac{1}{2} \underbrace{\vec{\omega}}_0 \cdot \frac{d\vec{L}}{dt} + \frac{1}{2} \frac{d\vec{\omega}}{dt} \cdot \vec{L} = 0$$

$$\Rightarrow \frac{d\vec{\omega}}{dt} \text{ is perpendicular to } \vec{L}, \quad \frac{d\vec{\omega}}{dt} \cdot \vec{L} = \sum_b \frac{d\vec{\omega}_b}{dt} L_b = 0$$

Then one can conclude: (change subindices a, b to i, j)

$$\boxed{\omega_i \frac{dI_{ij}}{dt} \omega_j = 0}$$

2) We know that:

$$(L_T^{CM})_a = \sum_{b=1}^3 I_{ab} \omega_b$$

→ Generically - moment inertia w.r.t CM. In our context it corresponds to the total moment of inertia since $\vec{r}_{CM} = 0$.

$$\omega_a (L_T)_a = \omega_a I_{ab} \omega_b \quad (\text{sum over } a \text{ and } b)$$

$$\frac{d\omega_a}{dt} (L_T)_a + \omega_a \frac{d(L_T)_a}{dt} = \frac{d\omega_a}{dt} I_{ab} \omega_b + \omega_a \frac{dI_{ab}}{dt} \omega_b + \omega_a I_{ab} \frac{d\omega_b}{dt}$$

* Since there is no net force or torque acting on the rigid body $\frac{d\vec{L}}{dt} = 0$, therefore $\left(\frac{d\vec{L}}{dt}\right)_a = 0$. But let us write out the equations for nonzero torque case.

We know from the previous question that $\omega_a \frac{dI_{ab}}{dt} \omega_b = 0$.

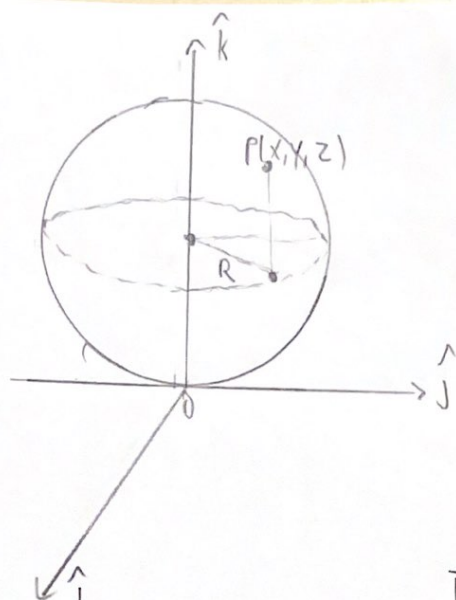
Also notice that $I_{ab} = I_{ba}$ since I is a symmetric matrix.

$$\left(\frac{d\omega_a}{dt}\right) (L_T)_a + \omega_a \frac{d(L_T)_a}{dt} = \left(\frac{d\omega_a}{dt}\right) \underbrace{I_{ab} \omega_b}_{(L_T)_a} + \frac{d\omega_b}{dt} \underbrace{I_{ba} \omega_b}_{(L_T)_b}$$

$$\frac{d\vec{\omega}}{dt} \cdot \vec{L}_T + \vec{\omega} \cdot \frac{d\vec{L}_T}{dt} = 2 \frac{d\vec{\omega}}{dt} \cdot \vec{L}_T$$

$$\boxed{\vec{\omega} \cdot \frac{d\vec{L}_T}{dt} = \frac{d\vec{\omega}}{dt} \cdot \vec{L}_T}$$

3)



For any point P inside the sphere

$$x^2 + y^2 + (z - R)^2 \leq R^2$$

$$I_{ab} = \sum_{i=1}^N [m_i p_i^2 \delta_{ab} - m_i (p_i)_a (p_i)_b]$$

Or in the case of continuous objects.

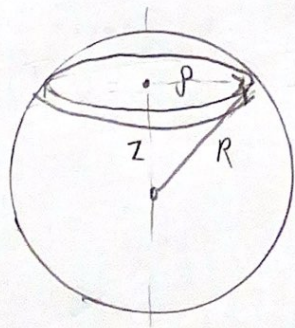
$$I_{ab} = \iiint_{xyz} \gamma dx dy dz [p_i^2 \delta_{ab} - (p_i)_a (p_i)_b]$$

γ is the density of the homogeneous sphere and its value is $\gamma = \frac{M}{\frac{4}{3}\pi R^3}$.

* Calculating the integrals involved is very labourous. We can instead use the Steiner's parallel axis theorem.

$$(I_{ij})^0 = (I_{ij})^{CM} + M r_{cm}^2 \delta_{ij} - M r_i (r_{cm})_i (r_{cm})_j$$

We can simply calculate $(I_{ij})^{CM}$ by just calculating I_{xx} . This is because there is a perfect symmetry in sphere it's symmetric about any axis that passes from its CM. Therefore $(I_{xx})^{CM} = (I_{yy})^{CM} = (I_{zz})^{CM}$ and $I_{xy} = 0, I_{xz} = 0, I_{yz} = 0$.



$$I^{CM} = \int_{-R}^R \int_0^{2\pi} \int_0^{\sqrt{R^2 - z^2}} p^2 \frac{M}{\frac{4}{3}\pi R^3} p d\theta dp dz = \frac{M}{\frac{4}{3}\pi R^3} \cdot 2\pi \int_{-R}^R \int_0^{\sqrt{R^2 - z^2}} p^3 dp dz$$

$$= \frac{3M}{2R^3} \int_{-R}^R \frac{(R^2 - z^2)^2}{4} dz = \frac{3M}{8R^3} \int_{-R}^R (R^4 - 2z^2 R^2 + z^4) dz$$

$$I^{CM} = \frac{3M}{8R^3} \left[R^4 z - \frac{2z^3 R^2}{3} - \frac{z^5}{5} \right]_{-R}^R$$

$$= \frac{3M}{8R^3} \left[2R^5 + \frac{2R^5}{5} - \frac{4R^5}{3} \right] = \frac{3M}{8R^3} R^5 \left(\frac{30+6-20}{15} \right) = \frac{2MR^5}{5}$$

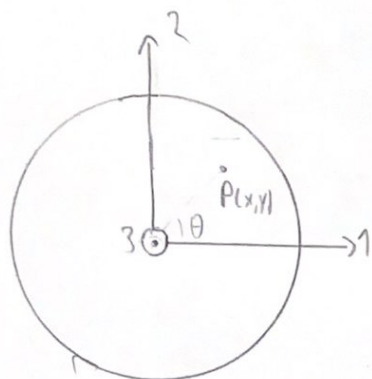
$$I_{ij}^{CM} = \begin{bmatrix} \frac{2}{5}MR^2 & 0 & 0 \\ 0 & \frac{2}{5}MR^2 & 0 \\ 0 & 0 & \frac{2}{5}MR^2 \end{bmatrix} ; M r_{cm}^2 \delta_{ij} = \begin{bmatrix} MR^2 & 0 & 0 \\ 0 & MR^2 & 0 \\ 0 & 0 & MR^2 \end{bmatrix} ; M(r_{cm})_i (r_{cm})_j = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & MR^2 \end{bmatrix}$$

$$(I_{ij})^0 = I_{ij}^{CM} + \begin{bmatrix} +MR^2 & 0 & 0 \\ 0 & MR^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

{ since $(r_{cm})_i = (r_{cm})_j = 0$
cross terms won't give
any contribution }

$$(I_{ij})^0 = \begin{bmatrix} \frac{7}{5}MR^2 & 0 & 0 \\ 0 & \frac{7}{5}MR^2 & 0 \\ 0 & 0 & \frac{2}{5}MR^2 \end{bmatrix}$$

4)

Assume $1 \rightarrow x$, $2 \rightarrow y$, $3 \rightarrow z$.

* Since axes shown are principal axes of the coin (they lie along symmetry axes) we don't need to worry about cross terms in moment of inertia. We just need to calculate I_{xx} , I_{yy} , I_{zz} or I_1 , I_2 , I_3

→ Notice $z=0$ for any point in the object since it's planar.

$$I_1 = \int_{-R}^R \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} \sigma (y^2 + z^2) dy dx \xrightarrow[\text{coordinates}]{\text{in polar}} I_1 = \int_0^R \int_0^{2\pi} \sigma \rho^2 \sin^2 \theta \rho d\rho d\theta$$

$$I_1 = \sigma \left[\int_0^R \rho^3 d\rho \right] \left[\int_0^{2\pi} \sin^2 \theta d\theta \right] = \sigma \frac{R^4}{4} \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{2\pi}$$

$$I_1 = \frac{\sigma \pi R^4}{4}$$

→ Due to symmetry I_2 also equals to $\frac{\sigma \pi R^4}{4} = I_2$

$$I_3 = \int_{-R}^R \int_{-\sqrt{R^2-y^2}}^{\sqrt{R^2-y^2}} \sigma (x^2 + y^2) dx dy = \int_0^R \int_0^{2\pi} \sigma \rho^2 \rho d\rho d\theta = \frac{\sigma \pi R^4}{2}$$

→ Clearly $I_3 = I_1 + I_2$

→ In fact the moment of inertia of the planar body about an axis perpendicular to the plane is equal to the sum of its moments of inertia about two perpendicular axes concurrent with perpendicular axis lying in the plane of the body

Perpendicular axis Theorem

5) Consider a falling cat either in free fall or in free ^{space}. In either case there can not be a torque due to gravity.

$$\vec{\tau}_{\text{net}} = \sum_{i=1}^N \vec{r}_i \times \vec{F}_i = \sum_{i=1}^N \vec{r}_i \times (m_i \vec{g}) = \left(\sum_{i=1}^N m_i \vec{r}_i \right) \times \vec{g} = 0$$

Then $\frac{d\vec{L}}{dt} = 0$ (air friction can be neglected)

* We can always choose a coordinate system, with its origin at CM point accelerating with $\vec{g}$.

$$\vec{L} = \vec{I} \cdot \vec{\omega} \quad \text{where} \quad \vec{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}$$

↳ there is a sum over required subindices so it represents a tensor product

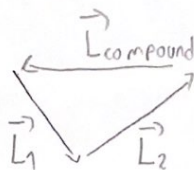
$$\frac{d\vec{L}}{dt} = \frac{d\vec{I}}{dt} \cdot \vec{\omega} + \vec{I} \cdot \frac{d\vec{\omega}}{dt} = 0 \quad \text{since angular momentum is conserved.}$$

Although initial angular velocity is zero, a cat can change its moment of inertia w.r.t its CM. So cat is not a rigid body

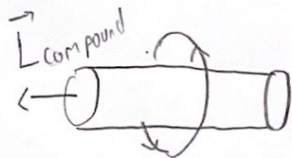
$$\frac{d\vec{I}}{dt} \neq 0 \Rightarrow \frac{d\vec{\omega}}{dt} \neq 0 \rightarrow \text{Therefore cat will rotate.}$$

Model For Falling Cat

Model the cat as concatenation of 2 cylinders. First cylinder represents it's upper body and the other it's lower body.



$$\vec{L}_{net} = 0$$



* Cat's can turn their upper and lower bodies separately. It give them advantage of rotating their upper/lower bodies in opposite directions while conserving angular momentum.

Initially cat draws in it's front paws (decreasing the moment of inertia of it's upper body). This way it's upper body is able turn more than it's lower body in the opposite direction. Then cat extends it's front paws and draws in it's back paws (decreasing the moment of inertia of it's lower body). This motion enables it to rotate his lower body greatly while it's lower body does not rotate as much. These sequence of rotating it's upper and lower body in opposite directions and adjusting their moment of inertia accordingly so that net angular momentum is zero allows cats to always fall on their feet.

6) Thornton & Marion 11.29

This is the case we've studied in class: $\left\{ \begin{array}{l} \text{Motion of the symmetric} \\ \text{top around } \theta=0 \end{array} \right.$

$$\dot{\phi} = \frac{P_{\phi} - P_{\psi} \cos \theta}{I \sin^2 \theta}$$

$$\dot{\psi} = \frac{P_{\psi}}{I_3} - \frac{(P_{\phi} - P_{\psi} \cos \theta) \cos \theta}{I \sin^2 \theta}$$

$$E' = \frac{1}{2} I \dot{\theta}^2 + \underbrace{\frac{(P_{\phi} - P_{\psi} \cos \theta)^2}{2I \sin^2 \theta}}_{V_{\text{eff}}(\theta)} + Mgh \cos \theta$$

where

$$P_{\phi} = (I \sin^2 \theta + I_3 \cos^2 \theta) \dot{\phi} + I_3 \cos \theta \dot{\psi}$$

$$P_{\psi} = I_3 (\dot{\psi} + \dot{\phi} \cos \theta) = I_3 \omega_3^0$$

$$E' = \frac{1}{2} I (\omega_1^2 + \omega_2^2) + Mgh \cos \theta$$

are constant of motion

$$\frac{V_{\text{eff}}}{I} = \tilde{V}_{\text{eff}}(\theta) = \frac{(a - b \cos \theta)^2}{\sin^2 \theta} + c \cos \theta \quad \text{where} \quad a \equiv \frac{P_{\phi}}{\sqrt{2}I}, \quad b \equiv \frac{P_{\psi}}{\sqrt{2}I}, \quad c \equiv \frac{Mgh}{I}$$

If we expand $\tilde{V}_{\text{eff}}(\theta)$ around $\theta=0$,

$$\tilde{V}_{\text{eff}}(\theta) = \frac{(a-b)^2}{\theta^2} + \left[\frac{1}{3} (a-b)(a+2b) + c \right] + \frac{1}{60} \theta^2 (4a^2 + 7ab + 4b^2 - 30c) + \dots$$

For $\theta=0$ not to be a singular point we need $a=b$

$$\left. \tilde{V}_{\text{eff}}(\theta) \right|_{a=b} = c + \underbrace{\frac{1}{60} \theta^2 (15b^2 - 30c)}_{\frac{1}{4} \theta^2 (b^2 - 2c)} + \frac{1}{24} \underbrace{(b^2 + c)}_{>0} \theta^4 + \dots$$

Since $b^2 > 0$, $c > 0$, we only need $b^2 > 2c$ to make sure that small perturbations does not grow unboundedly or $\theta=0$ is a stable point,

$$\hookrightarrow \frac{d\tilde{V}_{\text{eff}}}{d\theta} = 0, \quad \frac{d^2\tilde{V}_{\text{eff}}}{d\theta^2} > 0$$

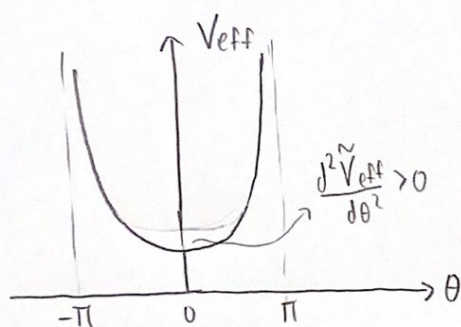
$$b^2 > 2c \quad \Leftrightarrow \quad \frac{p_\psi^2}{2I^2} > \frac{2Mgh}{I}$$

$$a = b \quad \Leftrightarrow \quad p_\phi = p_\psi \quad \Leftrightarrow \quad \dot{\phi}(0) = 0$$

$$p_\psi = I_3 (\dot{\psi} + \dot{\phi} \cos \theta) = I_3 \dot{\psi} = I_3 \omega_3$$

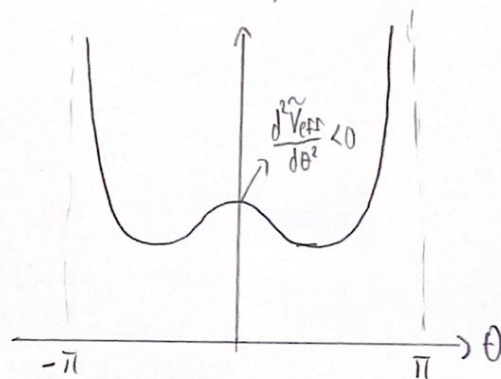
$$\text{Then our condition is } \frac{(I_3 \omega_3)^2}{2I^2} > \frac{2Mgh}{I}$$

* $\frac{4MghI}{I_3^2 \omega_3^2} < 1$ is required for stability



$$\frac{4MghI}{I_3^2 \omega_3^2} < 1, \quad b^2 > 2c$$

→ $\theta = 0$ is clearly stable
→ There are no other stable points



$$\frac{4MghI}{I_3^2 \omega_3^2} > 1, \quad b^2 < 2c$$

→ $\theta = 0$ is not stable
→ There are other stable points

* If the top is set off spinning with a sufficiently large ω_3 it will be stable. But as time goes on, due to friction (of fixed point or air friction) ω_3 will decrease. After ω_3 goes below the critical angular velocity $\omega_c = \frac{2}{I_3} \sqrt{MghI}$, spinning top will no longer be stable. It will start ^{doing} nutation before eventually dropping down.

This phenomenon is known as the sleeping top.

7) Thornton & Marion 11.30

$$\text{Eq. 11.62} \Rightarrow E' = \frac{1}{2} I_1 \dot{\theta}^2 + V(\theta)$$

$$E' = \frac{1}{2} I_1 \dot{\theta}^2 + \frac{(P_\phi - P_\psi \cos \theta)^2}{2 I_1 \sin^2 \theta} + Mgh \cos \theta$$

Letting $\dot{\theta} = 0$, one gets,

$$V(\theta) = \frac{(P_\phi - P_\psi \cos \theta)^2}{2 I_1 \underbrace{\sin^2 \theta}_{1 - \cos^2 \theta}} + Mgh \cos \theta = E'$$

$$P_\phi^2 - 2 P_\phi P_\psi \cos \theta + P_\psi^2 \cos^2 \theta + 2 I_1 Mgh \cos \theta - 2 I_1 Mgh \cos^3 \theta = 2 I_1 E' - 2 I_1 E' \cos^2 \theta$$

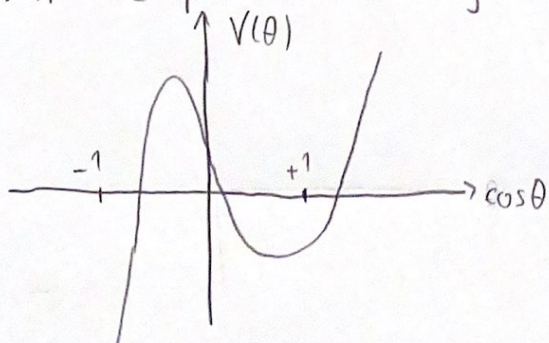
$$(2 I_1 Mgh) \cos^3 \theta - (P_\psi^2 + 2 I_1 E') \cos^2 \theta + 2 (P_\phi P_\psi - I_1 Mgh) \cos \theta + (2 I_1 E' - P_\phi^2) = 0$$

↳ This is clearly a cubic in terms of $\cos \theta$.

Let $A = I_1 Mgh$, $B = P_\psi$, $C = P_\phi$, $D = I_1 E'$

$$2 A \cos^3 \theta - (B^2 + 2 D) \cos^2 \theta + 2 (BC - A) \cos \theta + (2 D - C^2) = 0$$

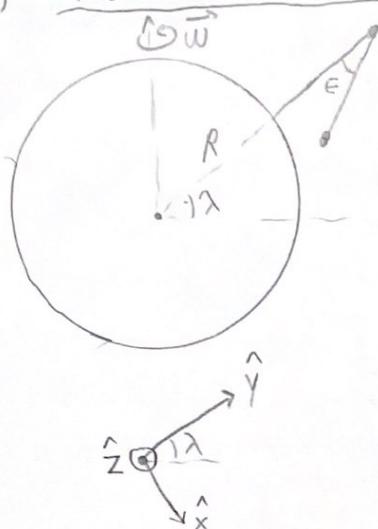
→ If we plot this using Mathematica:



→ Two of the roots lie in range $-1 \leq \cos \theta \leq 1$ and these correspond to real roots.

→ One root lies outside this range and it corresponds to.

8) Thornton & Marion 10.12



* Plumb line is essentially a mass hang from a string which is connected to a stable point. Assume system has reached a stable equilibrium (either we initialized it that way or reached a stable point after some damping) so that velocity of the mass w.r.t. Earth is 0. Stationary mass!
 $\vec{V}_{rot} = 0$

We know from our earlier discussions that:

$$\vec{a}_{rot} = \vec{a}_{in} - 2(\vec{\omega} \times \vec{V}_{rot}) - \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

OR

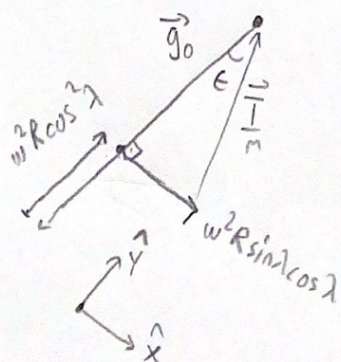
$$\vec{F}_{effective} = \vec{F}_{real} - \underbrace{2m(\vec{\omega} \times \vec{V}_{rot})}_0 - m\vec{\omega} \times (\vec{\omega} \times \vec{r})$$

Choose a coordinate system centered at origin. Also note that Earth's radius is much greater than height of the plumb line. So one can say $\vec{r} = R\hat{y}$ and $\vec{g}_0$ (real gravity) is still towards $-\hat{y}$ direction.

$$\vec{F}_{eff} = \vec{T} - mg_0\hat{y} - m(\omega \sin \lambda \hat{y} - \omega \cos \lambda \hat{x}) \times m(-\omega R \cos \lambda \hat{z})$$

$$\vec{F}_{eff} = \vec{T} - mg_0\hat{y} + m\omega^2 R \sin \lambda \cos \lambda \hat{x} + m\omega^2 R \cos^2 \lambda \hat{y}$$

→ Since mass is stationary $\vec{a}_{rot} = 0$ which means $\vec{F}_{eff} = 0$



$$\tan \epsilon = \frac{\omega^2 R \sin \lambda \cos \lambda}{g_0 - \omega^2 R \cos^2 \lambda}$$

For small angular deviation

$$\tan \epsilon \approx \epsilon = \frac{\omega^2 R \sin \lambda \cos \lambda}{g_0 - \omega^2 R \cos^2 \lambda}$$

Note that $g_0 - R\omega^2 \cos^2 \lambda$ is actually the effective gravity!
 You would weigh $R\omega^2 \cos^2 \lambda$ more if Earth was not rotating!
 And the term $R\omega^2 \sin \lambda \cos \lambda$ is the horizontal force acting on you due to centrifugal force.

$$\epsilon = \frac{R\omega^2 \sin \lambda \cos \lambda}{g_0 - R\omega^2 \cos^2 \lambda}$$

In order to find the latitude where maximum deviation occurs, take the derivative w.r.t. λ and equal it to zero.

$$\frac{d\epsilon}{d\lambda} = R\omega^2 \frac{(\cos^2 \lambda - \sin^2 \lambda)(g_0 - R\omega^2 \cos^2 \lambda) - 2R\omega^2 \cos^2 \lambda \sin^2 \lambda}{(g_0 - R\omega^2 \cos^2 \lambda)^2} = 0$$

$$\cos^2 \lambda g_0 - R\omega^2 \cos^4 \lambda - \sin^2 \lambda g_0 - R\omega^2 \cos^2 \lambda \sin^2 \lambda = 0$$

$$\cos^2 \lambda g_0 - \sin^2 \lambda g_0 - R\omega^2 \cos^2 \lambda = 0 \Rightarrow \tan^2 \lambda = \frac{g_0 - R\omega^2}{g_0} = 1 - \frac{R\omega^2}{g_0}$$

For Earth $\omega = 7.29 \times 10^{-5} \text{ rad/s}$, $R = 6.37 \times 10^6 \text{ m}$, $g = 9.89 \text{ m/s}^2$

$$\tan^2 \lambda = 0.9966 \approx 1 \Rightarrow \boxed{\lambda_{\text{max}} \approx 45^\circ}$$

$$\boxed{\epsilon_{\text{max}} \approx 0.002 \text{ rad}}$$